DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 3

Chatbot developer

Profession 3 of 10 in Development and IT.

What the profession means

A chatbot developer designs and builds conversations for websites, WhatsApp and messaging apps. Simple bots follow scripted intents; advanced bots use an AI model and the business's own documents, with a clear handover to a person.

What professionals do every day

Maps the conversations customers have, writes intents and answers, builds the bot in a builder or code, connects it to a CRM or calendar, tests with real questions and monitors the transcripts.

Why businesses hire them, and what problems they solve

Customers ask the same questions at any hour. A bot answers the common ones instantly and passes the rest to staff with the context already captured.

- Missed enquiries outside office hours.
- Staff answering the same questions again and again.
- Leads that go cold because the reply is slow.

Skills you need

- Conversation design: intents, slots, fallbacks and handover.
- Python or JavaScript for custom logic.
- Webhooks and APIs (Development and IT, Part 1).
- RAG for answers from documents (see Module D).
- Privacy, consent and opt-out for messaging channels.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Botpress	Visual chatbot builder	Free tier	Paid plans	botpress.com
Voiceflow	Design conversational flows	Free tier	Paid plans	www.voiceflow.com
Flowise	Open-source LLM flow builder	Open source	Paid cloud	flowiseai.com
WhatsApp Business Platform	Messaging channel for businesses	Test tools	Per-message pricing (check current)	developers.facebook.com/docs/whatsapp
Python	Scripts and logic	Free	Free	www.python.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	A rule-based FAQ bot with three intents and a handover.
Intermediate	A WhatsApp lead bot with a template reply and a calendar link.
Advanced	A knowledge assistant that answers from your documents with sources.

Projects by level

- **Beginner:** Website FAQ bot for a clinic.
- **Intermediate:** Lead-capture bot connected to a CRM.
- **Advanced:** Knowledge assistant over a product manual, with a monthly test set.

Client-ready portfolio project

Three chatbot projects (see Module D): a FAQ bot, a lead bot and a document assistant, each with a test log and a handover path.

Worked example

Worked example. A visitor asks 'how much is the course?'. The bot matches the word 'how much', returns the price answer, and asks which course. If nothing matches, it hands over with the captured question.

Assignment with full solution

ASSIGNMENT

A FAQ bot with a human handover

Business problem: A coaching centre wants a website chat that answers price, timing and booking questions, and hands everything else to a counsellor.

Steps

1. Write four intents (greeting, price, timing, booking) with keywords.
2. Add a fallback that hands over to a person.
3. Test with ten messages, including words that contain a keyword by accident (for example 'this' contains 'hi').
4. Record each unmatched message for review.

Expected output: A bot that passes the test set and a list of unmatched messages.

SOLUTION (ANSWER KEY)

Matching whole words avoids false matches such as 'hi' inside 'this'. Anything unmatched is handed over, so no question is silently ignored. The unmatched list is the best source for improving the bot.

PYTHON

```
import re

INTENTS = [
    ("greeting", ["hi", "hello", "hey"], "Hello! I can help with prices, timings and
bookings."),
    ("price", ["price", "cost", "fee", "fees", "how much"], "Our fees start at Rs 5,000. Tell
me the course and I will confirm."),
    ("timing", ["timing", "timings", "schedule", "when"], "Classes run on weekends. Which
batch do you prefer?"),
    ("booking", ["book", "enrol", "enroll", "join", "seat"], "Great! Please share your name
and phone number and a counsellor will call you."),
]
HANDOFF = "I will connect you with a counsellor for that."

def normalise(text):
    return re.sub(r"[^a-z ]", " ", text.lower())

def reply(message):
    words = normalise(message)
    for name, keywords, answer in INTENTS:
        for kw in keywords:
            if re.search(rf"\b{re.escape(kw)}\b", words):
                return name, answer
    return "handoff", HANDOFF

if __name__ == "__main__":
    assert reply("Hi there")[0] == "greeting"
    assert reply("how much is the course?")[0] == "price"
    assert reply("Can I book a seat for Monday?")[0] == "booking"
    assert reply("When are the timings?")[0] == "timing"
    assert reply("Is there a refund policy?")[0] == "handoff"
    assert reply("this is fine")[0] == "handoff"   # 'hi' inside 'this' must not match
    print("chatbot tests passed")
```

Common mistakes

- Matching partial words, so 'this' triggers the greeting.
- No handover, so questions are lost.
- Promising prices or dates the bot cannot verify.
- No monthly test set, so quality drifts unnoticed.

Freelancing opportunities

- Website and WhatsApp bots for small businesses.
- Maintenance and content updates for existing bots.
- Freelance briefs for 'chatbot', 'WhatsApp bot', 'Botpress', 'Voiceflow'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Botpress documentation
- Voiceflow documentation
- WhatsApp Business developer documentation

Part of **Master Freelancing: AI and Automation**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.